Midterm Examination – Project Management Principles and Practices (MS-PM)

Section A: Student Version (Total Marks: 30)

Time Allowed: 90 Minutes
Instructions:
- Attempt any *five* questions.
- All questions carry equal marks (6 each).
- Show complete workings for numerical questions.
- Use PMBOK 6th/7th edition terminology.

Q1. (Project Scope Management)

(a) Define *Project Scope Management* as per PMBOK.
(b) Explain the purpose and significance of a *Work Breakdown Structure (WBS).*
(c) Draw a three-level deliverable-based WBS for a project of your choice.
[Total: 6 Marks]

Q2. (Organizational Influences and PMO)

(a) Differentiate between *Enterprise Environmental Factors (EEF)* and *Organizational Process Assets (OPA).*
(b) Describe the main features of *Functional*, *Matrix*, and *Projectized* organizational structures.
(c) What are the three types of *Project Management Offices (PMO)* as defined in PMBOK?
[Total: 6 Marks]

Q3. (Project Life Cycles & Methodologies)

(a) Differentiate between *Predictive*, *Iterative*, and *Adaptive* life cycles.
(b) Identify one industry scenario best suited for each life cycle and justify your answer.
[Total: 6 Marks]

Q4. (Project Stakeholder Management)

(a) Define *Project Stakeholder* as per PMBOK.
(b) Explain the *Power–Interest Grid* and how it assists in stakeholder engagement strategy.
(c) Explain the purpose of a *Stakeholder Register* and list two components.
[Total: 6 Marks]

Q5. (Project Selection & Portfolio Management)

(a) Differentiate between *Project*, *Program*, and *Portfolio Management.*
(b) Explain any two *financial project selection techniques* (e.g., NPV, IRR, Payback).
(c) Describe the NPV decision rule as per PMBOK.
[Total: 6 Marks]

Q6. (Advanced PERT/CPM Numerical Problem)

The following data represent a network of **15 project activities (A1–A15)** with their immediate predecessors and time estimates (in weeks):

Activity	Predecessor(s)	Optimistic (to)	Most Likely (tm)	Pessimistic (tp)
A1	-	2	3	6
A2	A1	3	4	5
A3	A1	2	3	4
A4	A2	4	6	10
A5	A2	2	5	8
A6	A3	3	4	6
A7	A3	1	3	5
A8	A4,A5	3	5	9
A9	A6,A7	4	6	10
A10	A8	2	4	8
A11	A8	3	5	7
A12	A9,A10	2	3	4
A13	A9,A11	1	2	3
A14	A12,A13	2	4	6
A15	A14	1	3	5

Tasks:
1. Compute the *Expected Time (Te)* and *Variance (σ^2)* for each activity using:
$Te = (to + 4tm + tp)/6$ and $Variance = ((tp - to)/6)^2$ 2. Construct the *Network Diagram* (manually or digitally).
3. Identify the *Critical Path* and the *Expected Project Duration.*
4. Calculate the *Standard Deviation (σp)* of the project.
5. Determine the *probability of completing the project within 35 weeks* using Z-score:

$Z = (TargetTime - Te(total))/\sigma p$
(Use Standard Normal Table)
[Total: 6 Marks]

Section B: Faculty / Invigilator Version (Answer Key & Marking Scheme)

Q1 – Project Scope Management (6 Marks)

Detailed Solution Example for Students:
- *Definition:* Project Scope Management includes the processes required to ensure the project includes all the work required—and only the work required—to complete the project successfully (PMBOK 6th).
- *Purpose of WBS:* WBS helps break down the total project work into smaller, manageable components called work packages.
- *Example WBS (3 levels):*
- Level 1: Build Residential House
- Level 2: Design, Construction, Finishing
- Level 3 (Construction): Foundation, Structure, Roofing.
Tip: Remember the *100% rule*—WBS must cover 100% of the project scope.

Q2 – Organizational Influences (6 Marks)

Student Solution:
- *EEF:* External/internal conditions like market demand, legal environment, and organizational culture.
- *OPA:* Templates, processes, historical data, lessons learned repositories.
- *Structures:*
- Functional → Departmental control, weak PM authority (e.g., government organizations).
- Matrix → Shared control between PM and functional manager (e.g., Systems Ltd.).
- Projectized → PM has full authority (e.g., FWO).
- *PMO Types:*
- Supportive (advisory role),
- Controlling (enforces compliance),
- Directive (manages projects directly).

Q3 – Life Cycles (6 Marks)

Student Solution:
- *Predictive (Waterfall):* Requirements fixed, minimal change (e.g., highway construction).
- *Iterative:* Repeated cycles until final outcome known (e.g., research projects).
- *Adaptive (Agile):* Incremental delivery and feedback loops (e.g., app development).

Q4 – Stakeholder Management (6 Marks)

Student Solution:
- *Stakeholder Definition:* Any individual or organization affected by or can affect the project.

- *Power–Interest Grid:*
- High Power–High Interest → Manage Closely (Top Management).
- High Power–Low Interest → Keep Satisfied (Investors).
- Low Power–High Interest → Keep Informed (Users).
- Low Power–Low Interest → Monitor (Public).
- *Stakeholder Register:* Documents stakeholder information, influence, expectations, and communication needs.

Q5 – Portfolio Management (6 Marks)

Student Solution:
- *Project:* Temporary effort producing unique output.
- *Program:* Group of related projects (e.g., multiple IT upgrades).
- *Portfolio:* Collection of projects/programs aligned with strategic goals.
- *Financial Techniques:*
- *NPV:* Present value of inflows – outflows. Accept if >0.
- *IRR:* Discount rate at which NPV = 0.
- *Payback:* Time to recover investment.
Example: A project with an initial cost of $10,000 and annual inflows of $4,000 for 3 years → Payback = 2.5 years.

Q6 – Complete Step-by-Step PERT/CPM Solution for Students

Step 1: Compute Expected Time (Te) and Variance (σ^2)

Activity	to	tm	tp	Formula	Te	Variance
A1	2	3	6	(2+4×3+6)/6	3.5	0.44
A2	3	4	5	(3+4×4+5)/6	4.0	0.11
A3	2	3	4	(2+4×3+4)/6	3.0	0.11
A4	4	6	10	(4+4×6+10)/6	6.3	1.00
A5	2	5	8	(2+4×5+8)/6	5.0	1.00
A6	3	4	6	(3+4×4+6)/6	4.2	0.25
A7	1	3	5	(1+4×3+5)/6	3.0	0.44
A8	3	5	9	(3+4×5+9)/6	5.7	1.00
A9	4	6	10	(4+4×6+10)/6	6.3	1.00
A10	2	4	8	(2+4×4+8)/6	4.7	1.00
A11	3	5	7	(3+4×5+7)/6	5.0	0.44
A12	2	3	4	(2+4×3+4)/6	3.0	0.11

Activity	to	tm	tp	Formula	Te	Variance
A13	1	2	3	(1+4×2+3)/6	2.0	0.11
A14	2	4	6	(2+4×4+6)/6	4.0	0.44
A15	1	3	5	(1+4×3+5)/6	3.0	0.44

Step 2: Construct Network Diagram

A1 starts the project. A2 and A3 follow A1.
A4 & A5 follow A2; A6 & A7 follow A3.
A8 depends on A4 and A5.
A9 depends on A6 and A7.
A10 and A11 depend on A8.
A12 follows A9 & A10; A13 follows A9 & A11.
A14 depends on A12 & A13; A15 follows A14.

Network Flow Example:
A1 → A2 → A4 → A8 → A10 → A12 → A14 → A15
A1 → A2 → A5 → A8 → A11 → A13 → A14 → A15
A1 → A3 → A6 → A9 → A12 → A14 → A15
A1 → A3 → A7 → A9 → A13 → A14 → A15

Step 3: Identify Critical Path

From the Te values, longest expected path is:
A1 (3.5) + A2 (4.0) + A4 (6.3) + A8 (5.7) + A10 (4.7) + A12 (3.0) + A14 (4.0) + A15 (3.0)
→ **Total = 34.2 weeks (Critical Path)**

Step 4: Project Variance and Standard Deviation

Sum of variances for critical path:
0.44 + 0.11 + 1.00 + 1.00 + 1.00 + 0.11 + 0.44 + 0.44 = **4.54**
$\sqrt{}$4.54 = **2.13 weeks (σp)**

Step 5: Probability of Completion within 35 Weeks

$$Z = (Target - Expected)/\sigma p = (35 - 34.2)/2.13 = 0.38$$
From Z-table, P(Z<0.38) = 0.648 → **64.8% probability of finishing in 35 weeks.**

Step 6: Example Explanation

If management wants to complete the project within 35 weeks, the probability of success is about 65%.
If the deadline tightens to 33 weeks, the Z-score = (33 - 34.2)/2.13 = -0.56 → Probability = 0.2877 = 28.8%.
Hence, risk of delay increases significantly.

Step 7: Student Diagram Example

Students can draw a simple node-based diagram using boxes for activities, connecting arrows for dependencies, marking earliest and latest start times.

Example (simplified text version):

```
A1 → A2 → A4 → A8 → A10 → A12 → A14 → A15
 |     ↘ A5 ↗↘      ↗      A11
↘   A3 → A6 → A9↗             A13↗
```

This represents merge points at A8 and A9 leading into later activities.

End of Student Detailed Solution Section